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LEARNER GUIDE

For

Microsoft Certified Azure AI Engineer Associate AI-102 Training

TGS Ref No: TGS-2023036651

Conducted by
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Course Overview

This Learner Guide mirrors the lab sequence for the Microsoft Certified Azure AI Engineer Associate AI-102 Training course. Work through the labs in order and complete every validation check before moving on.

Course Code: TGS-2023036651

TSC Title: Artificial Intelligence Application in Product Development

TSC Code: ICT-TEM-4034-1.1

Before You Start

- Sign in to the Azure subscription or Skillable lab environment provided by the trainer.

- Create lab notes using the recommended files from the Tools Guide.

- Use the same naming convention for resources so cleanup is simple.

- Do not paste real confidential, personal, or customer data into lab prompts or documents.

Lab 01 - Plan, Manage, Monitor, and Secure an Azure AI Solution

Main Topic: Planning and operations

Scenario: A company wants a customer support AI platform with document search, answer generation, sentiment analysis, speech transcription, and image processing. You must plan the Azure AI resources.

Objectives

- Select appropriate Azure AI services.

- Plan resource deployment and endpoints.

- Identify authentication, keys, monitoring, and cost controls.

- Prepare an AI solution architecture.

Step-by-Step Lab Guide

Step 1: Map Requirements to Services

Requirement	Service
Generate responses	Azure OpenAI in Foundry Models
Search internal documents	Azure AI Search
Analyze sentiment	Azure AI Language
Transcribe calls	Azure AI Speech
Extract text from images	Azure AI Vision
Extract invoice fields	Azure AI Document Intelligence

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 2: Plan Foundry Resources

Document:

```
Hub or project name
Region
Model deployment
Default endpoint
Connected data source
Responsible AI settings
```

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 3: Plan Security

Record:

- Role-based access control.

- Managed identities.

- Key protection.

- Private endpoint or network controls.

- Logging and diagnostics.

- Secrets storage.

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 4: Plan Monitoring and Cost

Document:

```
Token or transaction usage
Latency
Errors
Content safety events
Budget alerts
Diagnostic logs
```

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 5: Draw Architecture

Use diagrams.net:

```
Application -> API layer -> Foundry/Azure AI services -> Search index/storage -> Monitoring
```

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Validation

You should have a service map, security plan, monitoring plan, and architecture diagram.

Checkpoint Questions

1. Why is service selection part of AI engineering?

2. Why should keys be protected?

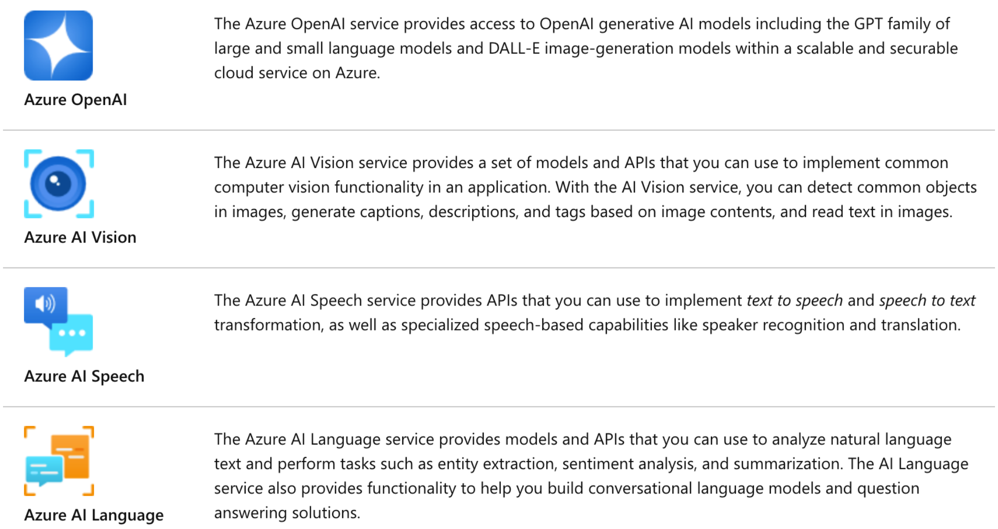
3. What should be monitored in an AI service?

4. What is the role of responsible AI planning?

Course Focus

AI engineering starts with selecting the right service, securing it, monitoring it, and planning for cost and responsible use.

Imported Reference Diagram



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Lab 02 - Responsible AI, Content Safety, Governance

Main Topic: Responsible AI

Scenario: The support AI assistant may receive abusive user input, produce inaccurate answers, or expose sensitive data. You must design responsible AI controls.

Objectives

- Apply responsible AI principles.

- Configure content safety concepts.

- Explain content filters, blocklists, prompt shields, and harm detection.

- Design a responsible AI governance framework.

Step-by-Step Lab Guide

Step 1: Review Responsible AI Principles

Write one control for:

```
Fairness
Reliability and safety
Privacy and security
Inclusiveness
Transparency
Accountability
```

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 2: Define Content Safety Controls

Document:

- Harm categories.

- Severity levels.

- Content filters.

- Blocklists.

- Prompt shields.

- Groundedness checks.

- Human escalation rules.

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 3: Create a Safety Policy

Write:

```
Allowed use:
Disallowed use:
Blocked content:
Human review trigger:
Audit log requirement:
Escalation owner:
Review frequency:
```

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 4: Test Governance Scenarios

Decide what should happen when:

- User asks for harmful instructions.

- Model produces low-confidence answer.

- Prompt contains personal data.

- Retrieved document is outdated.

- User asks for legal advice.

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 5: Create an Incident Plan

Define how to report, investigate, remediate, and prevent repeat AI safety incidents.

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Validation

You should have responsible AI controls, safety policy, scenario decisions, and incident plan.

Checkpoint Questions

1. What are content filters?

2. What is a prompt shield?

3. Why is human escalation important?

4. Who is accountable for AI output?

Course Focus

Responsible AI is implemented through policy, configuration, monitoring, human review, and governance.

Lab 03 - Generative AI, Foundry, Azure OpenAI, Prompt Flow, RAG

Main Topic: Generative AI

Scenario: The company wants a grounded assistant that answers questions from internal support articles and generates draft replies.

Objectives

- Plan a generative AI solution with Microsoft Foundry.

- Select and deploy a model conceptually.

- Design prompt templates and prompt flow.

- Implement a RAG pattern conceptually.

- Evaluate model and flow outputs.

Step-by-Step Lab Guide

Step 1: Select a Model

Record:

```
Model name
Use case fit
Context window
Latency
Cost
Safety settings
Deployment option
```

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 2: Design Prompt Templates

Create templates for:

- Direct answer.

- Answer with sources.

- Escalation when context is missing.

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 3: Plan Prompt Flow

Draw:

```
User question -> classify intent -> retrieve context -> prompt template -> model -> safety check -> response
```

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 4: Design RAG

Document:

- Source documents.

- Cleaning.

- Chunking.

- Embeddings.

- Vector store.

- Retrieval filters.

- Citation format.

- Evaluation method.

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 5: Evaluate Outputs

Score:

Criterion	Score 1-5
Correctness	
Grounding	
Helpfulness	
Safety	
Conciseness	

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Validation

You should have model selection notes, prompt templates, prompt flow diagram, RAG design, and evaluation table.

Checkpoint Questions

1. What is RAG?

2. Why are prompt templates useful?

3. What does tracing help debug?

4. How can feedback improve a generative AI solution?

Course Focus

Generative AI solutions need grounded data, evaluation, safety settings, monitoring, and feedback loops.

Imported Reference Diagram

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Lab 04 - Agentic AI, Foundry Agent Service, Multi-Agent Concepts

Main Topic: Agents

Scenario: The support AI assistant needs to look up order status, search support documents, draft a reply, and escalate complex cases to a human agent.

Objectives

- Explain agent roles and use cases.

- Plan Foundry Agent Service resources.

- Design custom agent tools and workflows.

- Review multi-agent orchestration concepts.

Step-by-Step Lab Guide

Step 1: Define Agent Use Case

Write:

```
Agent goal:
User types:
Allowed tools:
Disallowed actions:
Human escalation trigger:
Success metric:
```

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 2: Plan Agent Resources

Document:

- Foundry project.

- Model deployment.

- Search tool.

- Function or API tools.

- Authentication.

- Logging and trace settings.

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 3: Design Tool Calls

Create a table:

Tool	Purpose	Input	Output
Search knowledge base	Retrieve support articles	Query	Relevant passages
Order lookup	Get order status	Order ID	Status summary

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 4: Design Orchestration

Draw:

```
User request -> agent plan -> tool call -> observation -> model reasoning -> response ->
feedback
```

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 5: Multi-Agent Review

Explain when separate agents might be useful for retrieval, billing, technical support, and escalation review.

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Validation

You should have agent use case, resource plan, tool table, orchestration diagram, and multi-agent notes.

Checkpoint Questions

1. What is an AI agent?
2. Why do agents need tool permissions?
3. What is orchestration?
4. Why should autonomous actions be constrained?

Course Focus

Agentic solutions combine models, tools, workflows, monitoring, safety controls, and human escalation.

Lab 05 - Computer Vision, Custom Vision, Video Insights

Main Topic: Vision

Scenario: The company wants to inspect product photos, detect damaged items, read text from images, and extract insights from product demo videos.

Objectives

- Analyze images with Azure AI Vision concepts.

- Select visual features for image processing requirements.

- Compare image classification and object detection.

- Plan custom vision training.

- Explain video insight scenarios.

Step-by-Step Lab Guide

Step 1: Map Vision Requirements

Requirement	Vision Capability
Generate image tags	Image analysis
Detect product locations	Object detection
Read text from photos	OCR
Classify product condition	Image classification
Extract video topics	Video insights

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 2: Plan an Image Request

Document:

```
Image source
Visual features
Language
Expected response fields
Error handling
Privacy considerations
```

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 3: Custom Vision Plan

Choose:

Need	Model Type
One label for each image	Image classification
Locate multiple objects	Object detection

Document image labels, training images, evaluation metrics, publishing, and consumption.

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 4: Video Insights Plan

Explain how Video Indexer can extract:

- Transcript.

- Keywords.

- Faces or speakers where appropriate.

- Topics.

- Scenes.

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 5: Responsible AI Review

Document consent, retention, face analysis sensitivity, and human review needs.

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Validation

You should have vision mapping, request plan, custom vision plan, video insight notes, and responsible AI review.

Checkpoint Questions

1. What is OCR?

2. How is object detection different from classification?

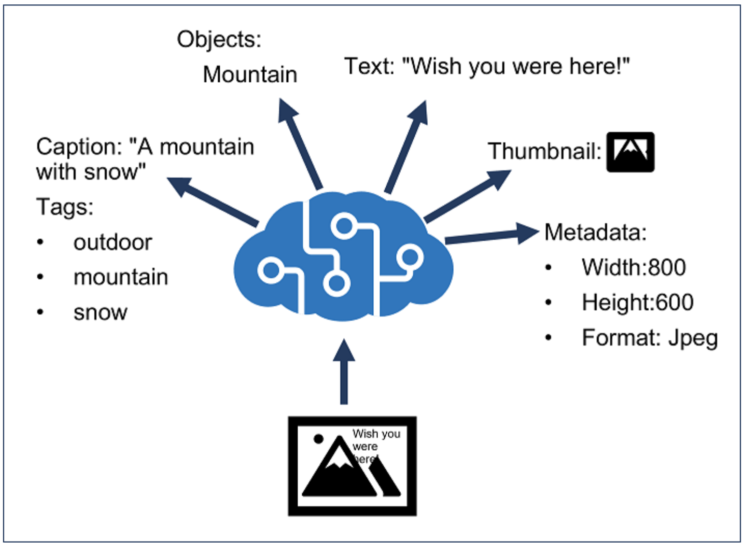
3. What does a custom vision model require?

4. Why is video analysis sensitive?

Course Focus

Vision solutions require selecting visual features, model type, evaluation approach, and responsible data handling.

Imported Reference Diagram



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Lab 06 - NLP, Language, Speech, Translation

Main Topic: Language and speech

Scenario: The support platform must analyze customer messages, detect PII, translate text, transcribe calls, and generate spoken responses.

Objectives

- Analyze and translate text.

- Extract key phrases, entities, PII, language, and sentiment.

- Process speech with speech-to-text and text-to-speech.

- Plan translation and SSML use.

Step-by-Step Lab Guide

Step 1: Map Language Tasks

Scenario	Capability
Identify customer sentiment	Sentiment analysis
Extract product names	Entity recognition
Identify important topics	Key phrase extraction
Detect personal data	PII detection
Detect message language	Language detection
Translate message	Translator

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 2: Design Text Analysis Pipeline

```
Input text -> language detection -> PII detection -> sentiment -> entities -> key phrases ->
route case
```

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 3: Design Speech Pipeline

```
Audio input -> speech to text -> intent or keyword recognition -> response -> text to speech
```

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 4: SSML Review

Explain how SSML can control voice, pitch, rate, pauses, and pronunciation.

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 5: Translation Review

Document:

- Text translation.

- Document translation.

- Speech-to-text translation.

- Speech-to-speech translation.

- Human review for critical messages.

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Validation

You should have language task mapping, text pipeline, speech pipeline, SSML notes, and translation notes.

Checkpoint Questions

1. What is entity recognition?

2. Why detect PII before storing text?

3. What is SSML?

4. When should translation outputs be reviewed?

Course Focus

Language and speech solutions combine analysis, privacy controls, translation, and application integration.

Imported Reference Diagram

Feature	Image
Speech to text	mcr.microsoft.com/product/azure-cognitive-services/speechservices/speech-to-text/about
Custom Speech to text	mcr.microsoft.com/product/azure-cognitive-services/speechservices/custom-speech-to-text/about
Neural Text to speech	mcr.microsoft.com/product/azure-cognitive-services/speechservices/neural-text-to-speech/about
Speech language detection	mcr.microsoft.com/product/azure-cognitive-services/speechservices/language-detection/about

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Lab 07 - Custom Language Models and Question Answering

Main Topic: Custom language

Scenario: The support assistant must recognize customer intents and answer frequently asked questions from approved sources.

Objectives

- Design custom language understanding.

- Define intents, entities, and utterances.

- Plan custom question answering.

- Create a multilingual question answering strategy.

Step-by-Step Lab Guide

Step 1: Define Intents and Entities

Create a table:

Intent	Example Utterance	Entities
CheckOrderStatus	Where is order 12345?	order number
RequestRefund	I want a refund for my laptop	product

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 2: Plan Training Data

Document:

- Number of example utterances.

- Balanced examples across intents.

- Entity labeling rules.

- Test set.

- Ambiguous utterances.

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 3: Plan Custom Question Answering

Record:

```
Knowledge sources
Q&A pairs
Alternate phrasing
Chit-chat
Multi-turn prompts
Publishing target
Fallback answer
```

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 4: Multilingual Strategy

Explain whether to translate sources, create separate projects, or translate user input before answering.

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 5: Backup and Recovery

Document export, versioning, backup, and recovery of language projects.

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Validation

You should have intent/entity table, training plan, question answering plan, multilingual strategy, and backup notes.

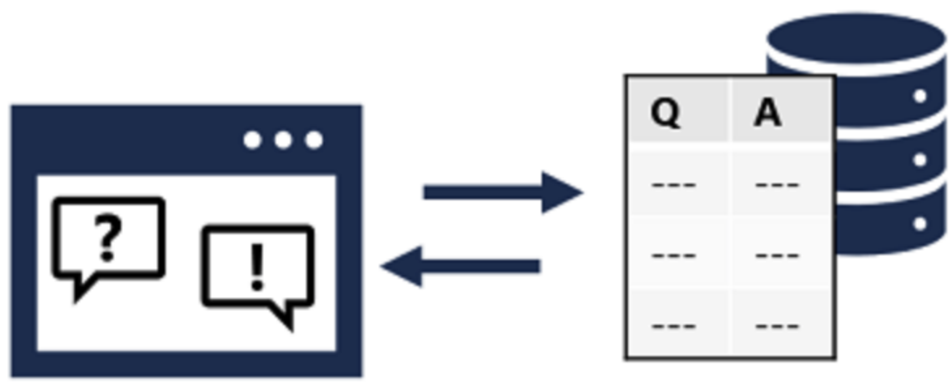
Checkpoint Questions

1. What is an intent?
2. What is an entity?
3. Why add alternate phrasing?
4. Why should language projects be backed up?

Course Focus

Custom language solutions need well-designed examples, evaluation, publishing, and lifecycle management.

Imported Reference Diagram



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Lab 08 - Azure AI Search, Knowledge Mining, Vector Search

Main Topic: Search and mining

Scenario: The company wants searchable support documents, enriched with extracted text, entities, key phrases, and vector embeddings for RAG.

Objectives

- Design an Azure AI Search solution.

- Define indexes, data sources, indexers, and skillsets.

- Explain semantic and vector search.

- Plan knowledge store projections.

Step-by-Step Lab Guide

Step 1: Design an Index

Define fields:

```
id
title
content
category
source_url
last_updated
security_label
embedding
```

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 2: Plan Data Sources and Indexers

Document:

- Source storage.

- Supported document formats.

- Indexer schedule.

- Change detection.

- Error handling.

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 3: Define a Skillset

Include:

- OCR.

- Language detection.

- Entity extraction.

- Key phrase extraction.

- Custom skill.

- Embedding generation.

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 4: Query Design

Write examples for:

```
Keyword search
Filter by category
Sort by last updated
Wildcard query
Semantic ranking
Vector search
Hybrid search
```

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 5: Knowledge Store Projections

Explain file, object, and table projections and when enriched output should be stored.

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Validation

You should have index design, data source/indexer plan, skillset plan, query examples, and knowledge store notes.

Checkpoint Questions

1. What is an indexer?

2. What is a skillset?

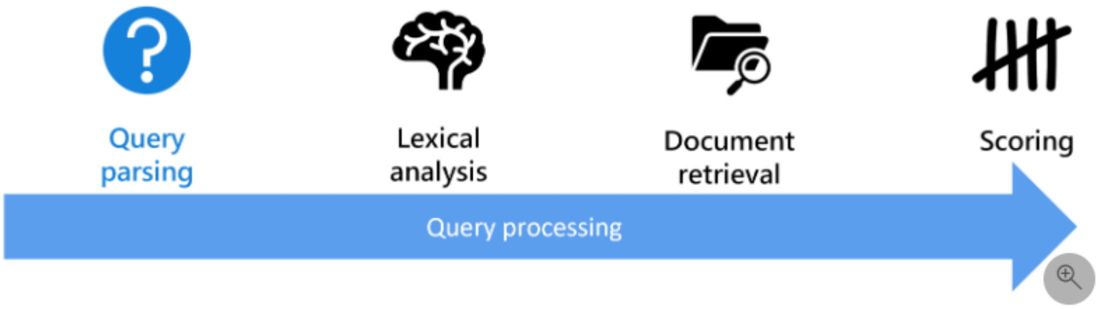
3. What is vector search?

4. Why use semantic ranking?

Course Focus

Knowledge mining combines source data, enrichment, indexing, querying, and optional vector retrieval for AI applications.

Imported Reference Diagram



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Lab 09 - Document Intelligence and Content Understanding

Main Topic: Document AI

Scenario: The company processes invoices, contracts, support screenshots, audio notes, and product videos. It wants to extract structured information from multiple content types.

Objectives

- Use prebuilt document extraction concepts.

- Plan custom Document Intelligence models.

- Explain composed models.

- Design Content Understanding extraction workflows.

Step-by-Step Lab Guide

Step 1: Map Document Scenarios

Scenario	Capability
Extract invoice fields	Prebuilt invoice model
Extract custom form fields	Custom document model
Route different form types	Composed model
Extract table data	Layout and table extraction
Extract text from images	OCR pipeline

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 2: Custom Document Model Plan

Document:

```
Document types
Sample count
Fields to extract
Labels
Training split
Evaluation metric
Publish plan
```

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 3: Content Understanding Plan

Explain extraction from:

- Documents.

- Images.

- Video.

- Audio.

Include summarization, classification, attributes, entities, tables, and images.

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 4: Human Review Rules

Define confidence thresholds and when extracted data requires review.

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 5: Integration Flow

Draw:

```
Upload content -> extract text/entities/tables -> validate -> store structured data ->
search/report
```

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Validation

You should have scenario mapping, custom model plan, content understanding plan, review rules, and integration flow.

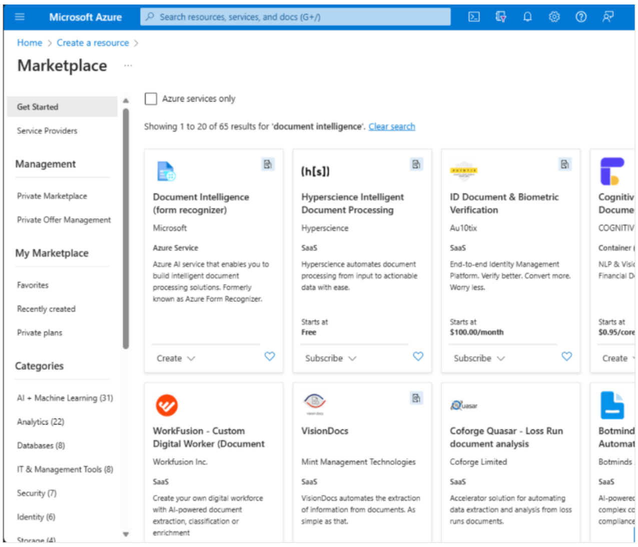
Checkpoint Questions

1. When should you use a prebuilt model?
2. What is a composed model?
3. Why are confidence scores useful?
4. What content types can Content Understanding process?

Course Focus

Information extraction solutions need model selection, training data, confidence handling, and integration design.

Imported Reference Diagram



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Lab 10 - AI-102 Capstone and Course Review

Main Topic: Integrated review

Scenario: You must design a production-ready customer support AI platform that supports grounded chat, document search, sentiment analysis, speech transcription, image OCR, document extraction, and human escalation.

Objectives

- Design an end-to-end Azure AI solution.

- Map requirements to Microsoft Foundry and Azure AI services.

- Include security, monitoring, responsible AI, and cost controls.

- Build a personal review plan.

Step-by-Step Lab Guide

Step 1: Build a Service Map

Requirement	Service
Grounded assistant	Foundry, Azure OpenAI, prompt flow, RAG
Agent workflow	Foundry Agent Service
Document search	Azure AI Search
Sentiment and PII	Azure AI Language
Speech transcription	Azure AI Speech
Image OCR	Azure AI Vision
Invoice extraction	Document Intelligence
Multi-content extraction	Content Understanding
Safety controls	Content Safety and responsible AI governance

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 2: Draw the Architecture

Include:

```
User application
API layer
Foundry project
Model deployment
AI Search index
Storage
Language/Speech/Vision/Document services
Monitoring
Human review queue
```

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 3: Add Governance Controls

Document:

- RBAC.

- Managed identity.

- Key protection.

- Prompt filters.

- Content safety.

- Logging and monitoring.

- Budget alerts.

- Data retention.

- Human escalation.

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 4: Rate Skill Confidence

Skill Area	Confidence 1-5	Study Action
Plan and manage Azure AI solution		
Implement generative AI solutions		
Implement agentic solution		
Implement computer vision solutions		
Implement NLP solutions		
Implement knowledge mining and information extraction		

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 5: Clean Up Resources

If you created resources:

```
az group delete --name rg-ai102-lab --yes --no-wait
```

Confirm with your instructor before deleting shared resources.

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Step 6: Create a 7-Day Review Plan

- Day 1: Planning, security, monitoring, responsible AI.

- Day 2: Generative AI, prompt flow, RAG.

- Day 3: Agents and orchestration.

- Day 4: Vision and video.

- Day 5: NLP, speech, translation, question answering.

- Day 6: AI Search, Document Intelligence, Content Understanding.

- Day 7: Capstone review and mistakes log.

1. Record the required outputs in your lab notes.
2. Ask the trainer to verify any uncertain configuration or design decision.

Validation

You should have a service map, architecture diagram, governance checklist, confidence matrix, cleanup plan, and review plan.

Checkpoint Questions

1. Which service should power grounded document search?
2. When should a human review AI output?
3. What is the difference between RAG and fine-tuning?
4. Which AI workload do you need to review most?

Course Focus

Azure AI engineering is about integrating the right AI services into secure, monitored, responsible, production-ready solutions.

Imported Reference Diagram

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Cleanup Checklist

- Delete training resource groups only when instructed by the trainer.

- Remove Foundry projects, model deployments, search indexes, storage accounts, and document processing resources created only for training.

- Confirm that no lab keys, endpoint values, or copied sample data remain in shared notes.

